

33-467: Astrophysics of Stars and The Galaxy

Week 1: logistics; observations & accessing software/data

logistics

- **syllabus overview**
- course philosophy & goals
- communication & textbooks
- semester schedule
- project overview

observations & accessing software/data

- astronomy as an empirical science
- sources of stellar data
- using observations to infer stellar properties
- using astronomical software
- principles of open source science

Department of Physics
Carnegie Mellon University

Course Outline

[33-467] Astrophysics of Stars and the Galaxy

Fall 2023

Instructor: Katelyn Breivik
E-mail: kbreivik@andrew.cmu.edu

Office: Wean 8319
Office Hours: 2pm on Mondays

Class Time: Tue/Thu 11-12:20

Class Room: DH 2105

course philosophy and goals

At the end of the class, students should be able to:

- ☐ formulate a comprehensive approach to answering open questions in astronomy that includes using physical principles and astronomical data to support their arguments.
- ☐ present research through oral presentations and text format with supporting figures.
- ☐ compute stellar properties from observations and relate the present state of a star to its state at any time in the future.
- ☐ describe how a stellar companion can alter the evolution of stellar populations.
- ☐ understand how stars form and be able to demonstrate how the interstellar medium interacts with the initial and end states of stars.
- ☐ be conversant with the current state of research in extrasolar planets and understand how our solar system is different to and similar to other detected solar systems.
- ☐ understand the structure of the Milky Way galaxy, and able to relate the dark matter distribution to that of stars, gas, dust and black holes.

Course Component	Overall Weight	Tentative Period
Individual Homework		
<i>Assignments (8 total)</i>	20%	Week 2-14
Projects		
<i>Project 1: mid-project update</i>	10%	Week 4
<i>Project 1: final presentation</i>	30%	Week 7
<i>Project 2: mid-project update</i>	10%	Week 12
<i>Project 2: final presentation</i>	30%	Week 15

Final grade boundaries are: A: 90 - 100% B: 80 - 89% C: 70 - 79% D: 60 - 69% R: <60%
 Boundary cases may be discussed based on demonstrated engagement, effort and improvement over the semester.

attendance and late work

I won't be keeping attendance, but you are more likely to do well if you come to class for lectures and project time

Homework will be accepted up to one week after the due date, but will be reduced by 20%

course philosophy and goals

Everyone in this course is must be treated with respect. Science is a human endeavour and we are all accountable for ensuring that the environment we create is welcoming and free from discrimination. Each of us is here for a variety of reasons, comes from a variety of backgrounds, and have a diversity of skills and past experiences. It is expected that the students and instructor will contribute to a collaborative learning environment where diverse perspectives and approaches are celebrated.

see syllabus for advice on reporting incidents of hostility on the basis of identity as well as resources for student wellness including the CMU food pantry

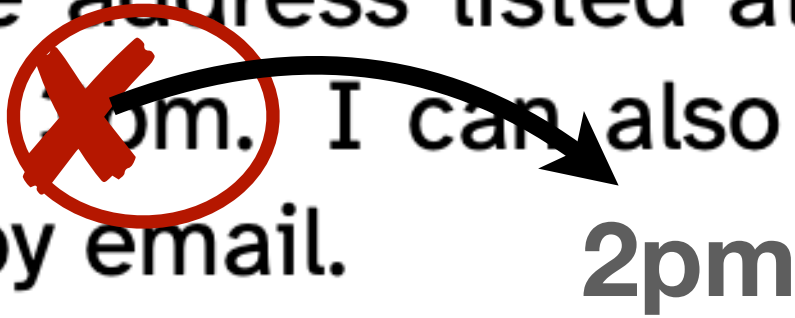
If you have accommodations or think you would benefit from accommodations, let me know asap so I can work with you!

communication and textbooks

Textbooks and Course Materials

-  **Canvas:** The site can be reached from <http://www.cmu.edu/canvas/> The course website will contain the syllabus, homework, project assignments and all required reading. Announcements and visual material presented during lectures will also be posted throughout the semester.
-  **Textbooks:** There is no required textbook for this course. All required reading will be available through Canvas. The majority of the stellar structure and evolution material will be drawn from Onno Pols' [lecture notes](#).

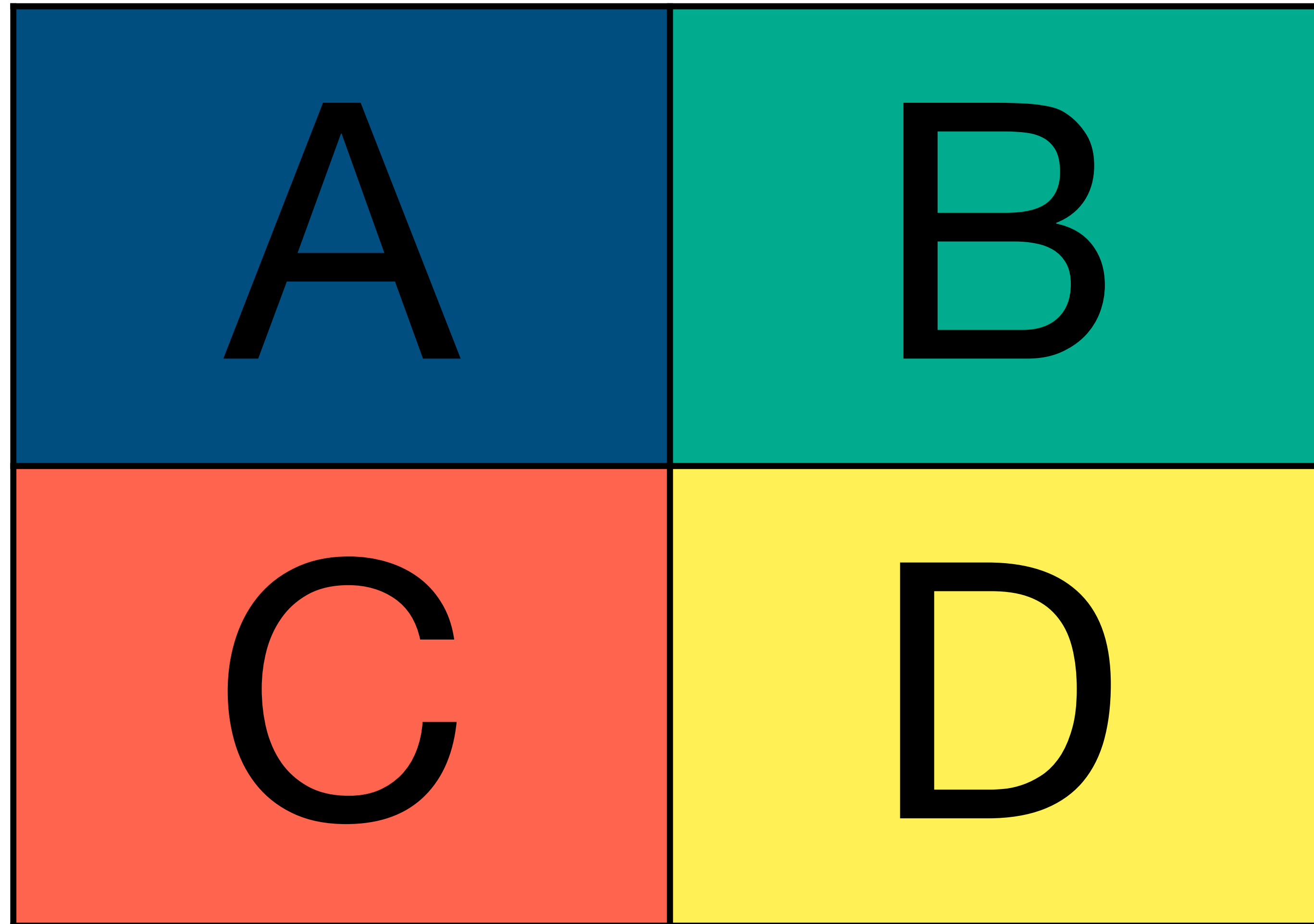
communication and textbooks

The majority of our communication will be done during classtime but I am also reachable by email at the address listed at the top of this page. I will also hold a weekly office hour on Mondays at ~~1pm~~  2pm. I can also meet outside of those hours, but these meetings will need to be scheduled by email.

We will also be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, our grader, and myself. Rather than emailing questions directly to me, I encourage you to post your questions on Piazza. If you run into any issues, please reach out to me so we can get them sorted as soon as possible! Find our class page at:

<https://piazza.com/cmu/fall2023/33467a/info>

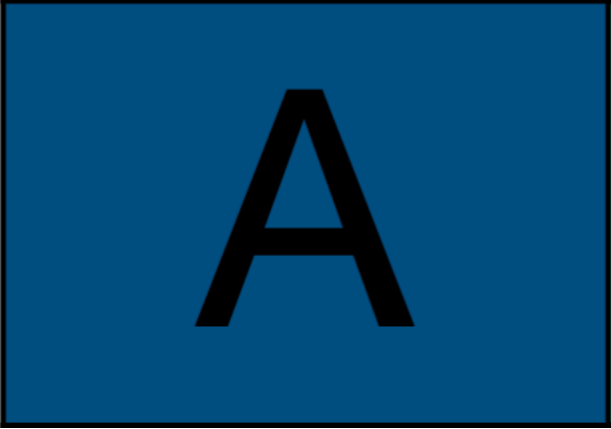
communication and textbooks



planned schedule

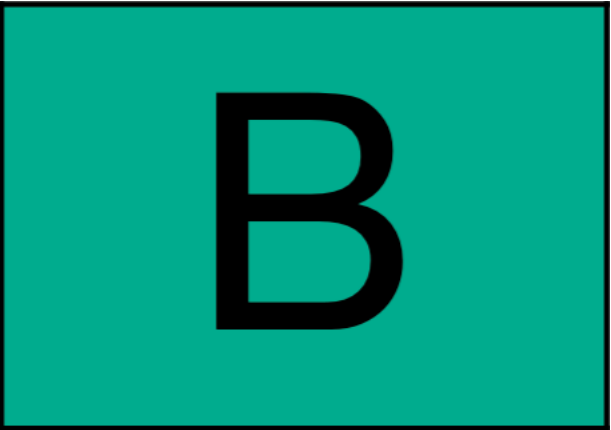
Week	Topics & Homework
1	Standard Observations, Accessing and Processing Data
2	Equilibrium, Timescales, and Stellar Interiors Project #1 Selection: 09/05 Assignment:# 1: 09/07
3	Stellar Interiors and Energy Transport Assignment:# 2: 09/14
4	Energy Transport and Nuclear Processes Project #1 Mid-way presentations
5	Nuclear Processes and Low-mass Stars Assignment:# 3: 09/26
6	Massive Stars and Stellar Remnants Assignment:# 4: 10/05
7	Project #1 Final presentations
8	Fall Break

First color vibe!

A solid blue square with a black border, containing the white capital letter 'A' in the center.

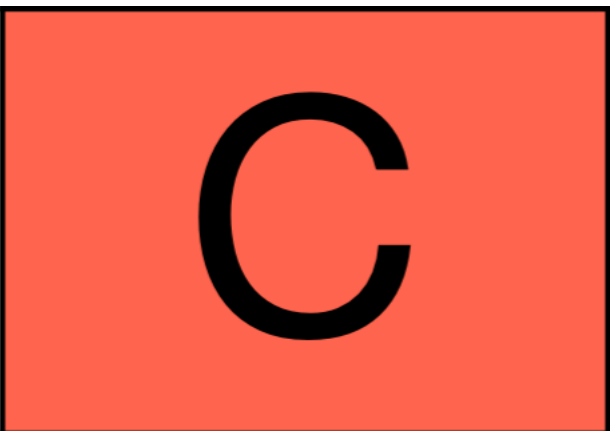
A

I strongly prefer in-class mid-project presentations

A solid teal square with a black border, containing the white capital letter 'B' in the center.

B

I strongly prefer scheduled zoom mid-project presentations

A solid red square with a black border, containing the white capital letter 'C' in the center.

C

I do not have a strong preference

What's the deal with projects?

1	The Fate of the Earth, and other Exoplanets	1
2	Testing Stellar Models with Wide Binaries	1
2.1	Calibrating the Initial to Final Mass Relation for White Dwarfs	2
2.2	Testing Stellar Age Measurements from Isochrones	2
3	Analyzing Periodic Signals with Lightkurve	2
3.1	Measuring the Rotation Rates of Stars	2
3.2	Estimating Stellar Properties with Asteroseismology	2
4	Measuring Star Cluster Ages	2
5	Designing Surveys for the Stars	3
6	Supernova Natal Kicks	3
7	The Black Hole Mass Function	3

Projects are listed on Canvas Page

What's the deal with projects?

You are allowed (encouraged!) to team up with a partner if you choose

Multiple groups (up to 2) can work on the same project, but you will need to talk with each other to coordinate final presentations!

Project selection will be done by first come, first serve on Piazza by posting your project and partner in the project_1 selection folder

The first project must be selected by next Tuesday — 09/05/23

Final questions about logistics?

Astronomy is empirical by construction

We have one Universe to observe — no repeats!

The combination of theory, computation, and data analysis is what allows us to make sense of our observations

Astronomy is *relative* by construction

Every measurement is made with respect to a *reference*

In this course, we will often work with **solar** quantities

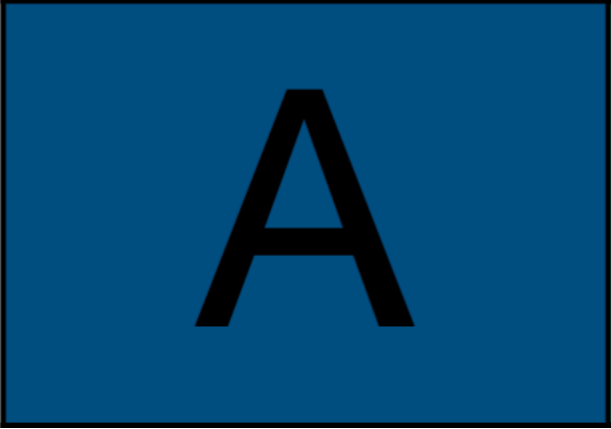
$M_{\odot}, R_{\odot}, L_{\odot}$ — why tho?

Why “stars” and “the Galaxy”

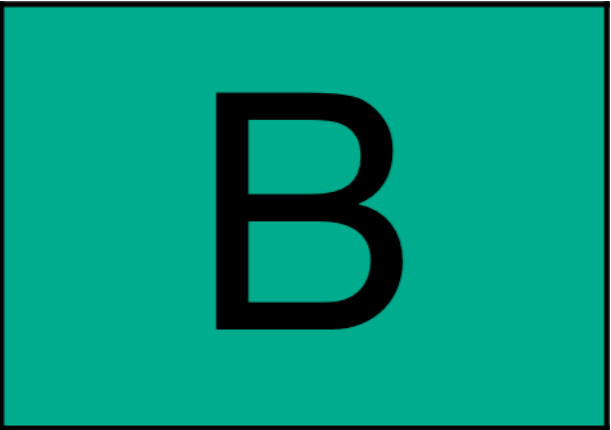
Stars are the building blocks of galaxies,
and therefore,
the **Universe**

The Milky Way (and local group) is the only galaxy (group)
that we can study in **immense detail**

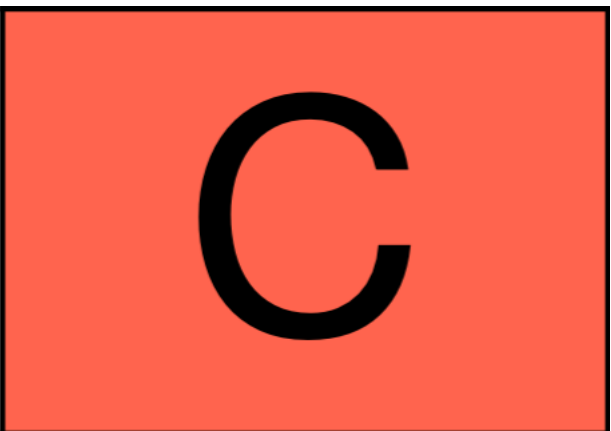
Color vibe time!

A dark blue square with a black border containing the white capital letter 'A'.

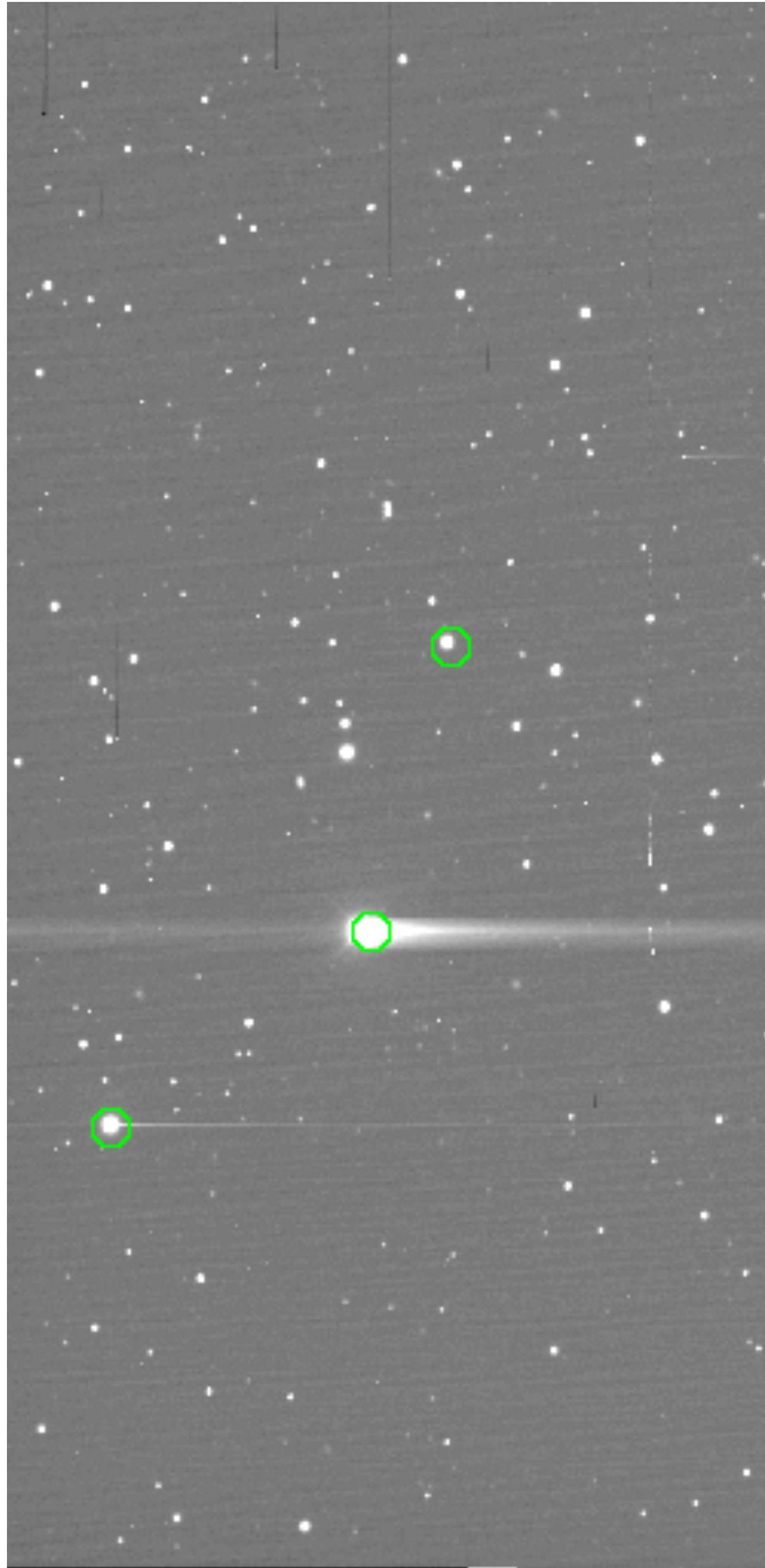
I am 100% on board with “why stars and the Galaxy”

A teal square with a black border containing the white capital letter 'B'.

I’m mostly on board but want to learn more

A red square with a black border containing the white capital letter 'C'.

I want to talk more about “why stars & the Galaxy”



Sources of stellar data: Photometry

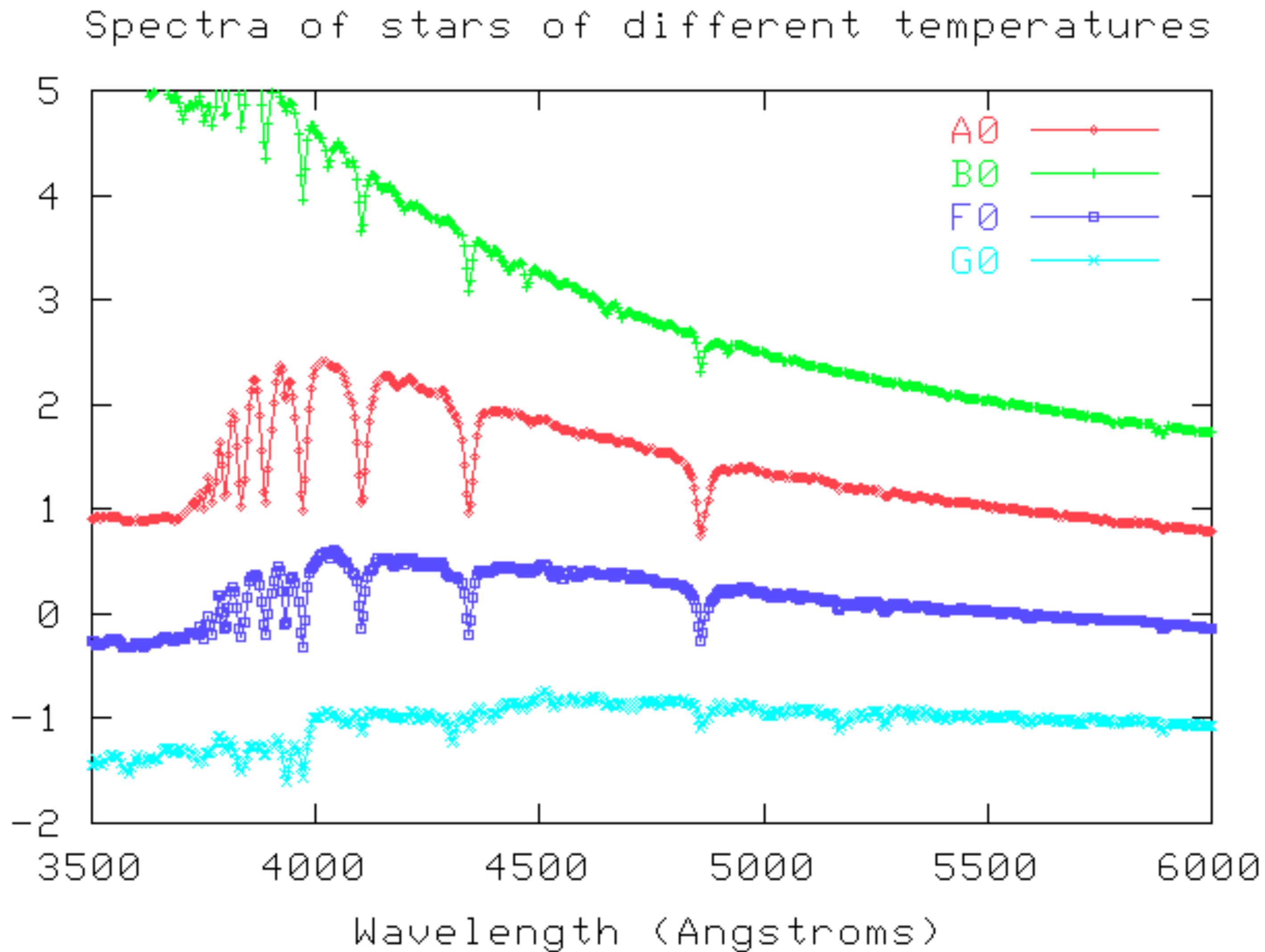
Photometry yields the *apparent* brightness (magnitude), or energy flux received at Earth, in different “colors” defined by filters with different bandpasses

For empirical reasons only: low magnitudes means brighter stars

The flux in different colors can be used to infer stellar *temperatures*

Bolometric corrections can be created from stellar models & data and used to infer the apparent bolometric flux

Sources of stellar data: Spectra



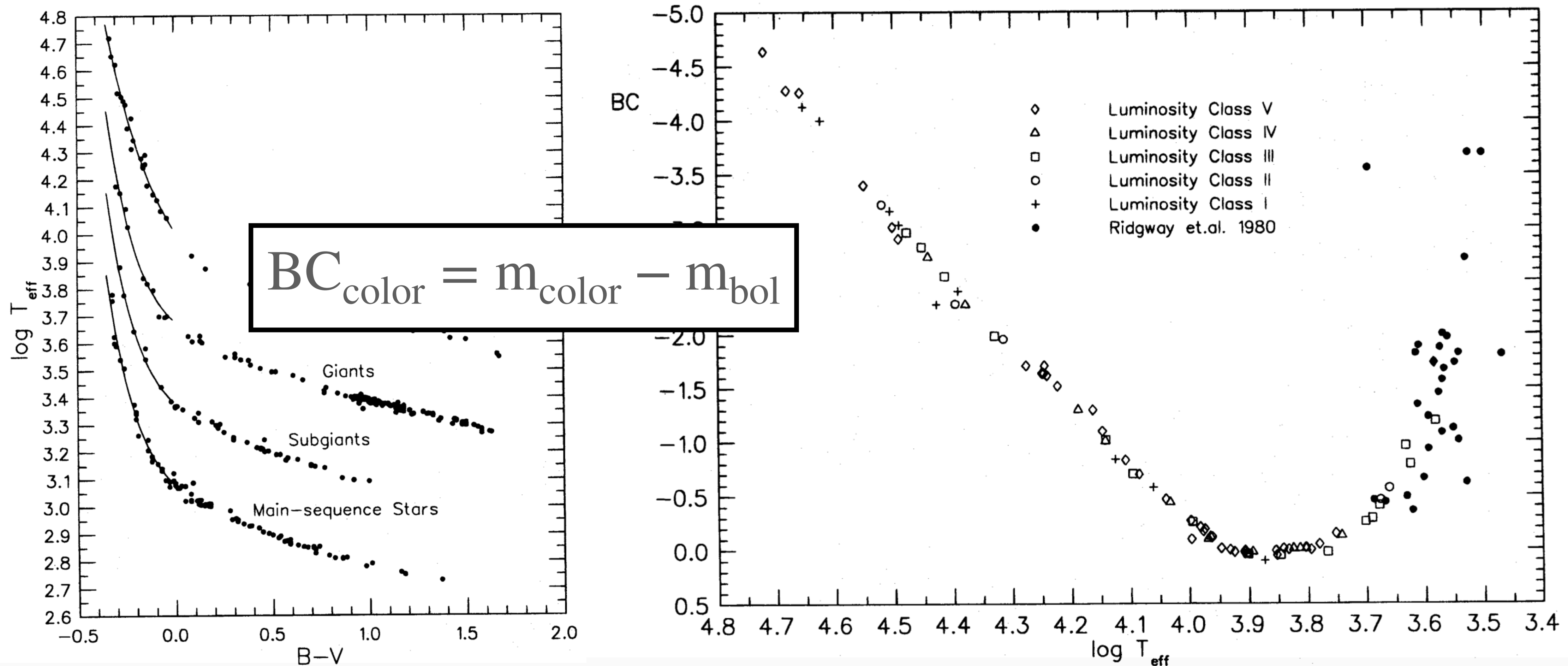
Spectra measure the flux at different wavelengths and can be used to infer the **temperature** and *chemical composition* of stars

$$\lambda_{\text{max}} = \frac{0.0029 \text{ m K}}{T_{\text{eff}}}$$

Sources of stellar data: Photometry

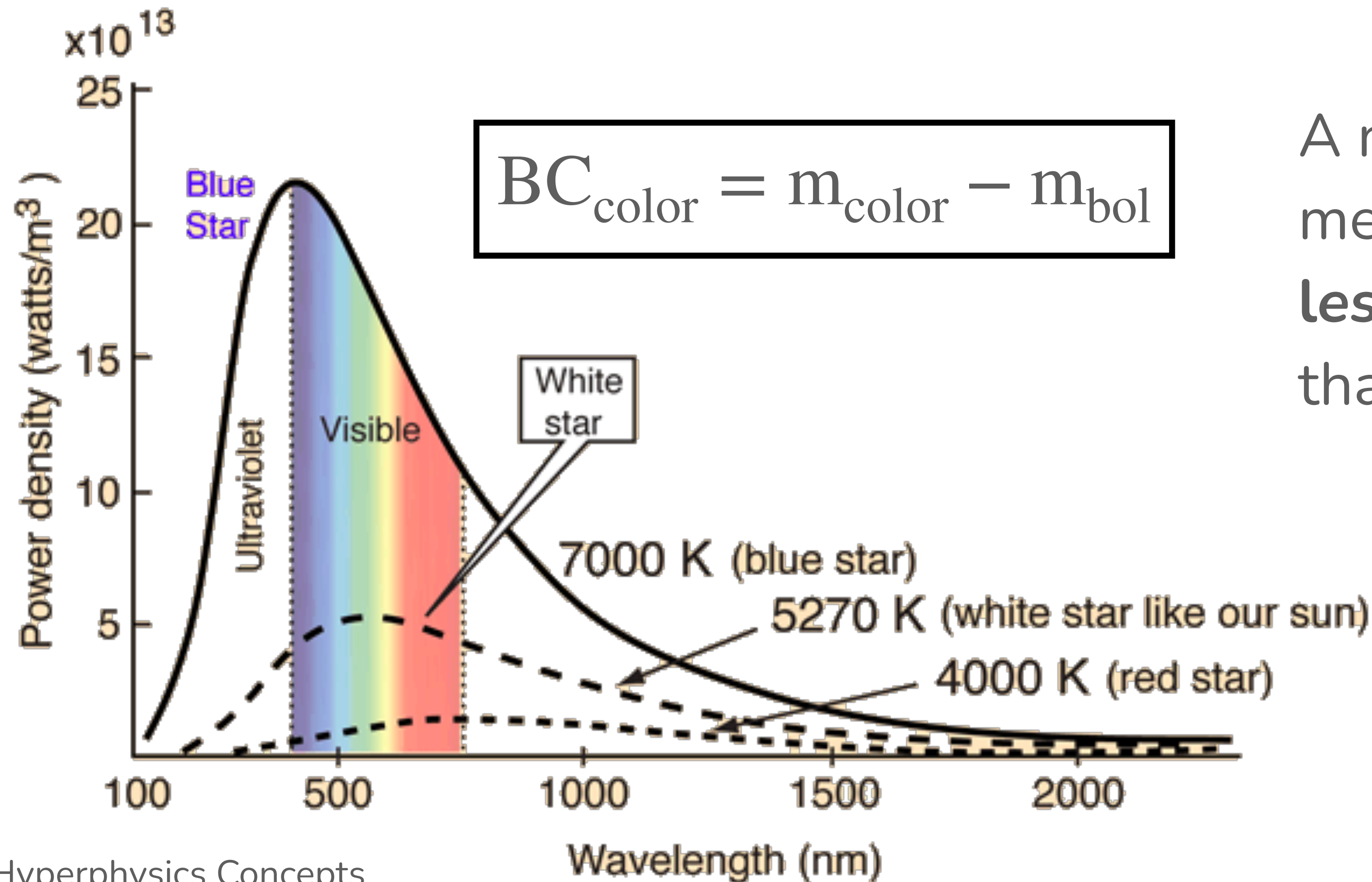
Bolometric corrections can be created from stellar models & reference data and used to infer the bolometric flux

Flower 1996



Sources of stellar data: Photometry

Stars emit blackbody radiation —
based on their temperature, they emit light at different wavelengths.

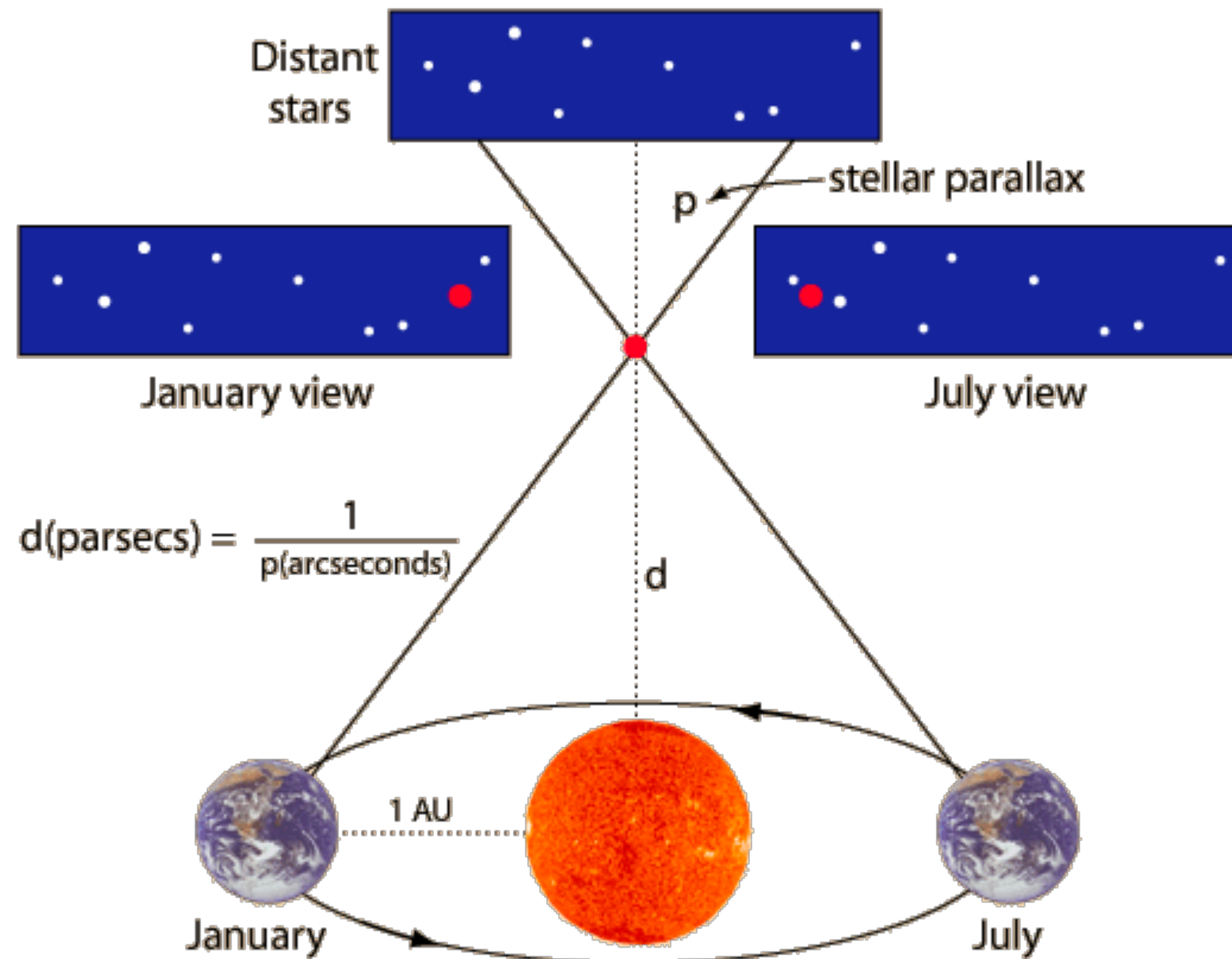


$$BC_{\text{color}} = m_{\text{color}} - m_{\text{bol}}$$

A negative BC for a blue filter means that the star emits ***less light at hot temperatures*** than the reference star

Sources of stellar data: Photometry

The ***absolute*** magnitude of a star, or its absolute brightness, can only be determined if we know how far away it is.



Parallax is the simplest way to measure distances



Gaia has measured parallaxes for almost 2 billion stars

Sources of stellar data: Photometry

The *absolute magnitude* of a star, or its absolute brightness, can only be determined if we know how far away it is.

$$m_{\text{app}} - M_{\text{abs}} = 5 \log_{10}(d/10 \text{ pc})$$

$$M_{\text{abs,bol}} = M_{\odot,\text{abs,bol}} - 2.5 \log_{10}(L/L_{\odot})$$

Sources of stellar data: Photometry

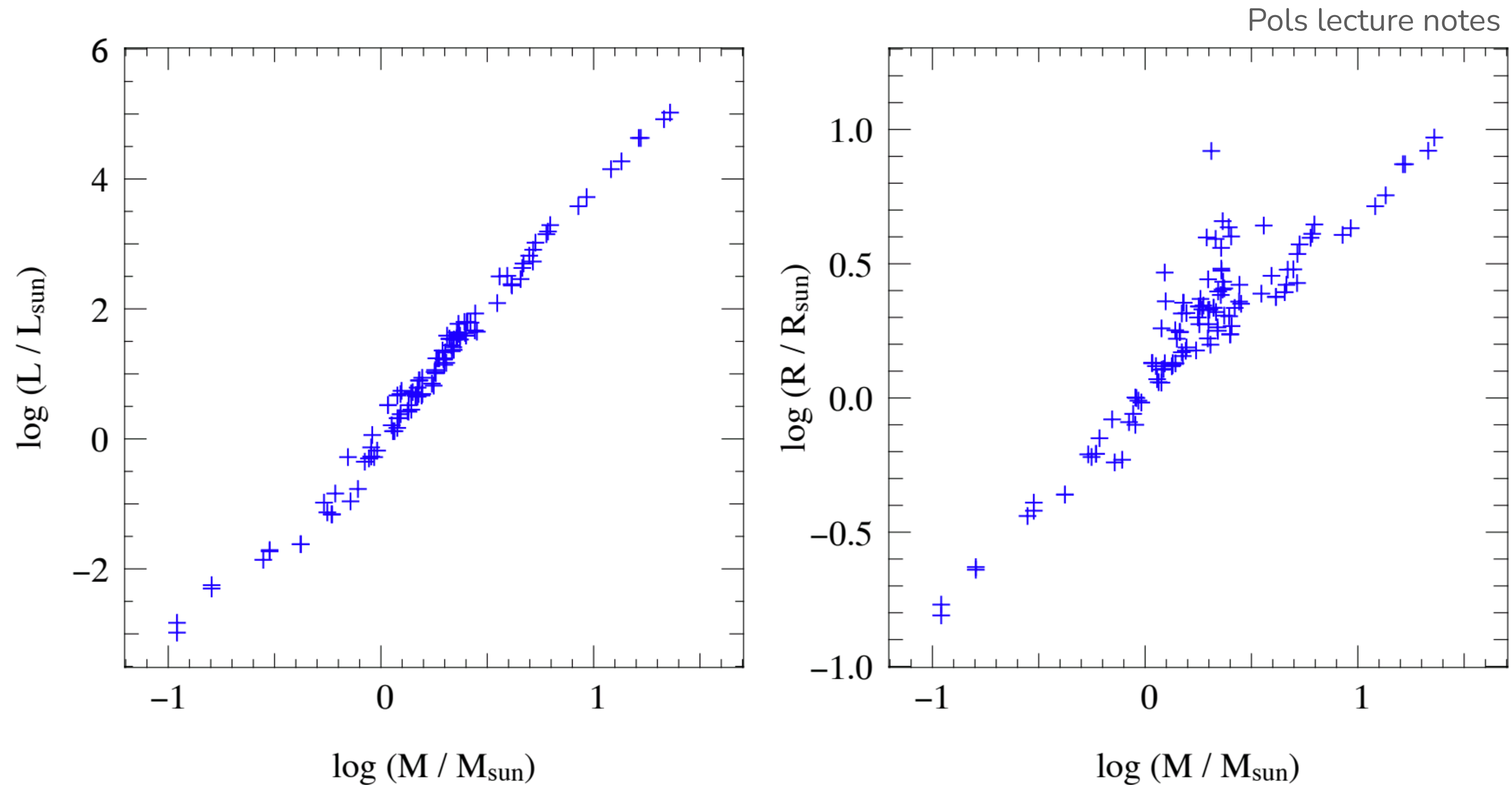
The *luminosity* of a star and its **temperature** can be used to infer the **radius** of the star using the Stefan-Boltzmann equation

$$L = 4\pi R^2 \sigma_{\text{SB}} T_{\text{eff}}^4$$

σ_{SB} : Stefan-Boltzmann constant

Mass of a single star is extremely difficult to
measure without the use of
stellar modeling or orbital dynamics

Main Sequence masses verified by “double-line eclipsing binaries”



$$(L/L_{\odot}) \approx (M/M_{\odot})^{3.8}$$

$$(R/R_{\odot}) \approx (M/M_{\odot})^{0.7}$$

Let's talk about how we actually
do astronomy

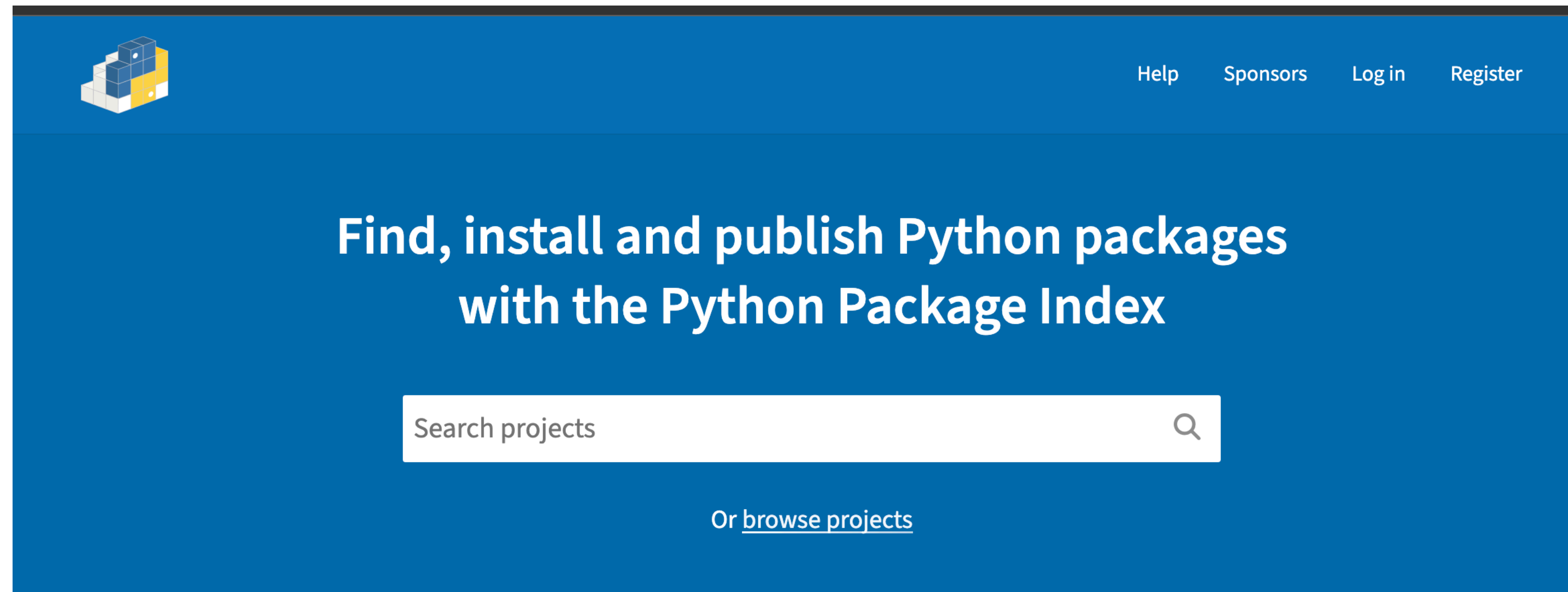
Most astronomy research can be done using
Python.

Stellar astrophysics uses Fortran because it is
lightning fast and a lot of stellar astrophysics was
worked out in the 1970s-90s
(they didn't have Julia/JAX back then!)

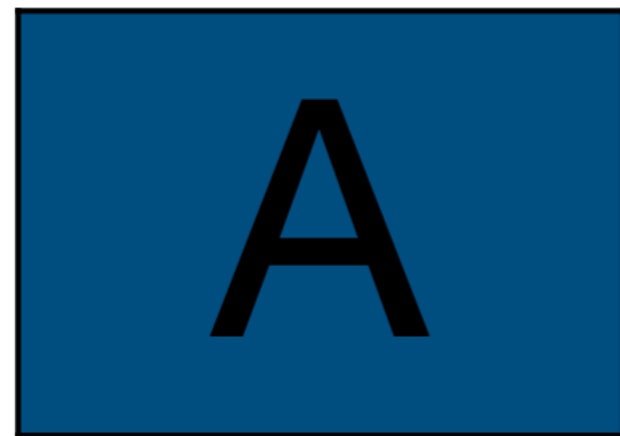
Python packages are almost always
community-developed (open source)



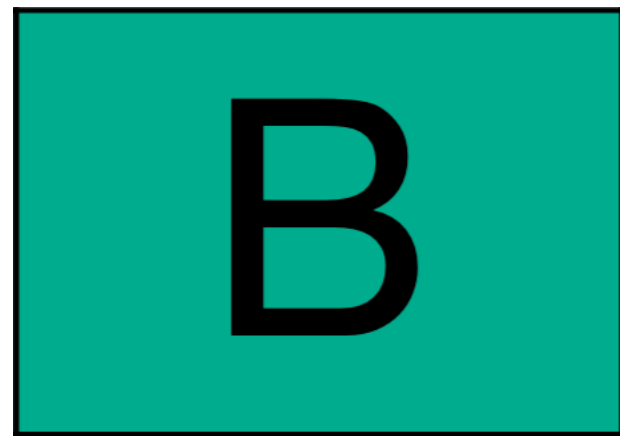
You can access python code using the Python Package Index (PyPI - or pip)



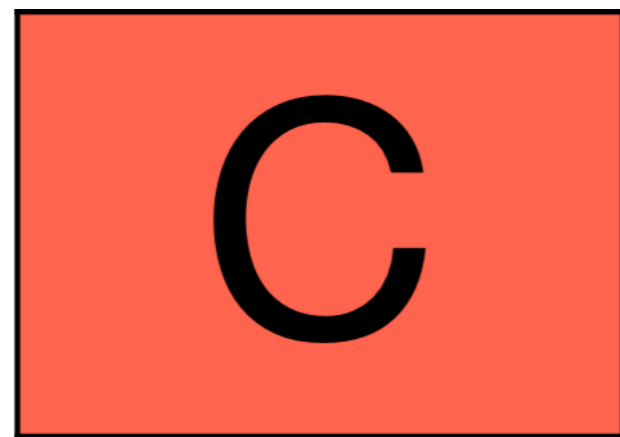
Color vibe time!



I am comfortable with using and installing Python code



I have used Python before but could use some help



I have never used Python before

Instructor promise:

I will help you work through your software problems in class, in office hours, and on Piazza

A major goal of this course is for everyone to be able to show an  on that color vibe question

Instructor expectation:

You will help your classmates when you can!

Many of you will run into similar problems and will be able to explain fixes to each other better than I can because you share common knowledge and experiences

As an open-source user & developer, there are some principles that I follow

1. There is no good or bad code.
 - 1.1. There is open or closed development
 - 1.2. There is code that works & code that doesn't
2. If you use someone else's code, you ***MUST*** give them credit
 - 2.1. This is usually through citations or shout-out's in presentations
3. Before you contact a developer, you should read the documentation and error messaging
 - 3.1. This means you need to be able to articulate a question specifically about the docs and/or the error!
4. Anyone who uses open-source code can ***also develop*** open-source code
 - 4.1. Most code is on Github — you can submit issues if you find bugs — you can submit pull-requests if you make a fix! (I vow to help you learn this skill if you are interested and find bugs in your projects!)

